

Technical Architecture Addendum: Target Technology Stack

Document Track: Core Systems Architecture | System Target: Project Aegis Distributed CAS Engine

1. STACK TOPOLOGY & COMPONENT SELECTION

Layer / Subsystem	Primary Technology	Multi-Region Target	Architectural Rationale
Edge & Ingress Routing	Cloudflare Workers / Rust Envoy	AWS CloudFront + Lambda@Edge	Hardware-accelerated Anycast routing with native support for HTTP/3 over QUIC [span_0] (start_span)[span_0](end_span) . Offloads HMAC validation directly at edge PoPs before traffic hits the ingestion core [span_1] (start_span)[span_1](end_span) .
Ingestion Engine & FastCDC	Rust (Tokio + SIMD fastcdc)	Go (gcflags optimized)	Zero garbage collection pauses during large streaming payloads; memory-safe rolling buffer execution via SIMD instruction sets (AVX-512 / ARM NEON) [span_2] (start_span)[span_2](end_span) .
Metadata Control Plane (Transactional)	PostgreSQL (v16+) on NVMe + <code>ltree</code>	Google Cloud Spanner / CockroachDB	Native GiST index support for <code>ltree</code> hierarchies, transactional stored procedures, and low transaction latency (<5 ms) for localized deployments [span_3] (start_span)[span_3](end_span) .
Fast Index & Leases	Redis Cluster (or DragonflyDB)	AWS ElastiCache	In-memory chunk existence pre-filtering, upload session tracking, and distributed lock management via fencing tokens [span_4] (start_span)[span_4](end_span) .
CDC & Event Streaming	Redpanda	Apache Kafka + Tiered Storage	High-throughput metadata append log for downstream derivation workers and asynchronous compaction event pipelines [span_5] (start_span)[span_5](end_span) .
CAS Binary Data Plane	Ceph / MinIO / AWS S3 / Cloudflare R2	Backblaze B2 / Custom NVMe JBOD	Hardware-level erasure coding (e.g., 8+4 RS), cold/hot storage tiering, and direct pre-signed PUT workflows that bypass compute runtimes entirely [span_6] (start_span)[span_6](end_span) .
Workflow Orchestration	Temporal.io + Go/Rust Workers	AWS Step Functions + SQS	Manages retries, state machine deduplication, and long-running distributed tasks (ClamAV scans, OCR, batch GC) [span_7] (start_span)[span_7](end_span) .

2. COMPONENT DESIGN & OPERATIONAL MECHANICS

Edge & Ingress Layer

- Routes inbound traffic across Anycast points of presence[span_8](start_span)[span_8](end_span).
- Evaluates ephemeral, tenant-scoped HMAC signatures directly at the edge layer, dropping unauthorized payload transfers prior to backend compute ingress[span_9](start_span)[span_9](end_span).

Data Processing & Chunking Plane (Rust)

- Implements the FastCDC algorithm via an asynchronous stream consumer to process streaming byte buffers dynamically[span_10](start_span)[span_10](end_span).
- Utilizes normalized target boundaries (1 MB target, 64 KB min, 4 MB max) to eliminate boundary shifts and deduplicate modified byte streams[span_11](start_span)[span_11](end_span).

Relational Control Plane

- Materializes graph lineage paths via `ltree` representations to avoid recursive graph traversals on directory mutations[span_12](start_span)[span_12](end_span).
- Executes directory moves and cyclic ancestor checks within single-statement ACID transactions using pessimistic row-level locking[span_13](start_span)[span_13](end_span).

Event Sourcing & Derivation Fleet

- Emits Change Data Capture (CDC) events on successful version commits to streaming topics (Redpanda/Kafka)[span_14](start_span)[span_14](end_span).
- Routes derivation tasks out-of-band to asynchronous worker fleets, isolating CPU-intensive workloads from the core upload hot path[span_15](start_span)[span_15](end_span).

3. RECOMMENDED ARCHITECTURAL ADJUSTMENTS

- **Decoupled Global CAS Registry:** Remove `tenant_id` from the primary key of the CAS registry to maximize storage savings across all shared blocks, managing tenant isolation and authorization exclusively within the manifest layer[span_16](start_span)[span_16](end_span).
- **Distributed Bloom Filter Garbage Collection:** Replace relational table anti-joins with partitioned Bloom filter sweeps or external MapReduce passes across storage inventory logs to safely sweep unreferenced objects without locking the primary database[span_17](start_span)[span_17](end_span).
- **Optimistic / Fenced Version Commits:** Enforce explicit row locks or version sequence epochs on parent nodes during manifest commits to prevent race conditions during concurrent file modifications[span_18](start_span)[span_18](end_span).